

Year 12
Mathematics Extension 2
HSC Trial Examination
2014

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen. Black pen is preferred.
- Board-approved calculators may be used
- A table of standard integrals is provided on the back page of this paper
- In questions 11 – 16, show all relevant reasoning and/or calculations

Total marks – 100

Section I

10 marks

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II

90 marks

- Attempt Questions 11–16
- Allow about 2 hours and 45 minutes for this section

DO NOT REMOVE THIS PAPER FROM THE EXAMINATION ROOM

Section I

10 marks

Attempt Questions 1 – 10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Question 1 – 10.

- 1** An object moving in a circular path of radius 4 metres travels 48 metres in 3 seconds.
The angular speed of the object is:
- (A) 3 rad/s
- (B) 4 rad/s
- (C) 12 rad/s
- (D) 16 rad/s
- 2** What is the gradient of the tangent to the circle $x^2 - 2x + y^2 = 9$ at the point $(0, -3)$?
- (A) $-\frac{1}{3}$
- (B) $-\frac{11}{6}$
- (C) $\frac{1}{3}$
- (D) $\frac{1}{6}$
- 3** The number of ways that 6 items can be divided between 3 people so that each person receives 2 items is:
- (A) 6
- (B) 27
- (C) 90
- (D) 360

4 Which of the following is the expression for $\int \sin^3 x dx$?

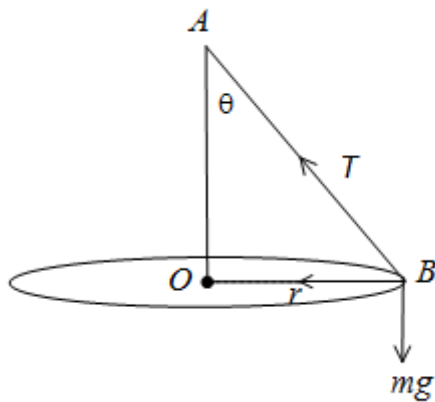
(A) $\frac{1}{3} \cos^3 x - \cos x + C$

(B) $\frac{1}{3} \cos^3 x + \cos x + C$

(C) $\frac{1}{3} \sin^3 x - \sin x + C$

(D) $\frac{1}{3} \sin^3 x + \sin x + C$

5 A particle at B is attached to a string AB that is fixed at A . The particle rotates in a horizontal circle with a radius of r . Let T be the tension in the string and $\angle BOA = \theta$.



Which of the following statements is correct?

(A) $T \cos \theta - mg = ma$

(B) $T \sin \theta = mr\omega^2$

(C) $T = -mg$

(D) $T - mg = ma$

- 6** The polynomial equation $3x^3 - 2x^2 + x - 7 = 0$ has roots α, β and γ .

Which polynomial equation has roots $\frac{2}{\alpha}, \frac{2}{\beta}$ and $\frac{2}{\gamma}$?

(A) $3x^3 - 4x^2 + 4x - 56 = 0$

(B) $7x^3 - 2x^2 + 8x - 24 = 0$

(C) $x^3 - 2x^2 - 27x - 49 = 0$

(D) $24x^3 - 8x^2 + 2x - 7 = 0$

- 7** The point $P\left(cp, \frac{c}{p}\right)$ lies on the rectangular hyperbola $xy = c^2$. The equation of the normal to the hyperbola at P is:

(A) $x + pqy = c(p + q)$

(B) $x + p^2y = 2cp$

(C) $px - \frac{1}{p}y = cp^2\left(1 - \frac{1}{p^2}\right)$

(D) $py - c = p^3(x - cp)$

- 8** What is the eccentricity for the hyperbola $\frac{y^2}{225} - \frac{x^2}{64} = 1$?

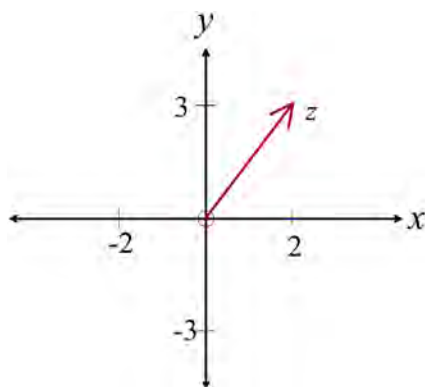
(A) $\frac{8}{17}$

(B) $\frac{15}{17}$

(C) $\frac{17}{15}$

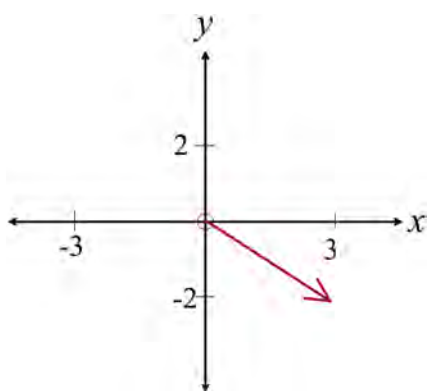
(D) $\frac{17}{8}$

- 9 The Argand diagram below shows the complex number z .

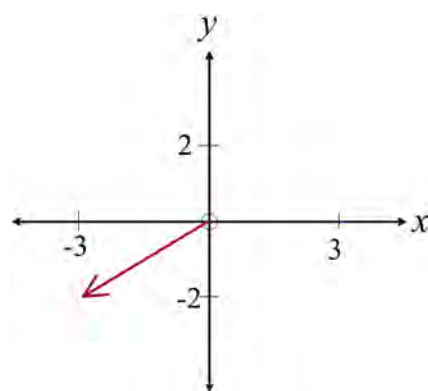


Which Argand diagram best represents $i\bar{z}$?

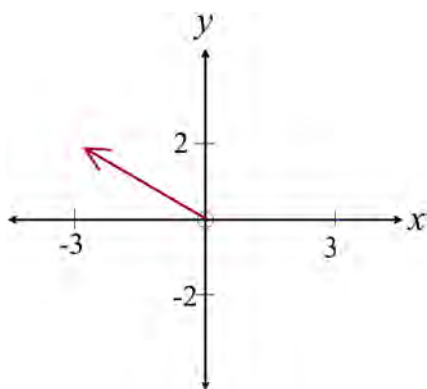
(A)



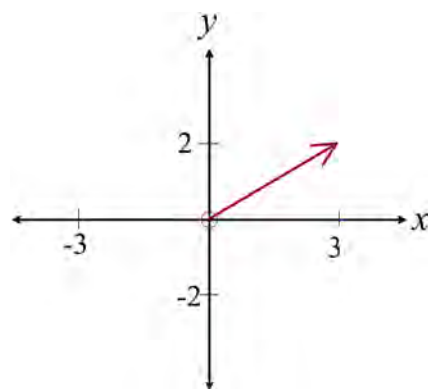
(B)



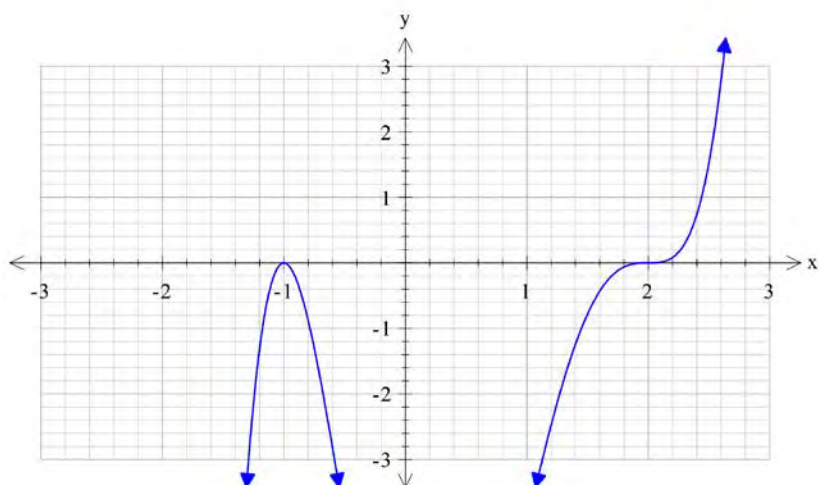
(C)



(D)

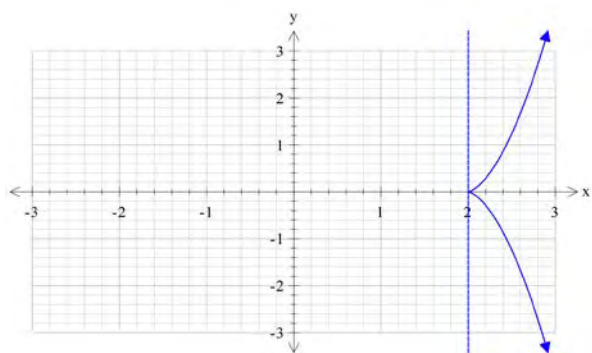


10 The graph of $y = f(x)$ is drawn below.

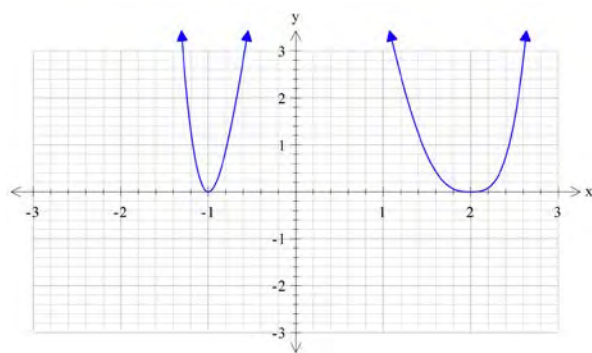


Which of the following graphs represents the graph of $y = [f(x)]^2$?

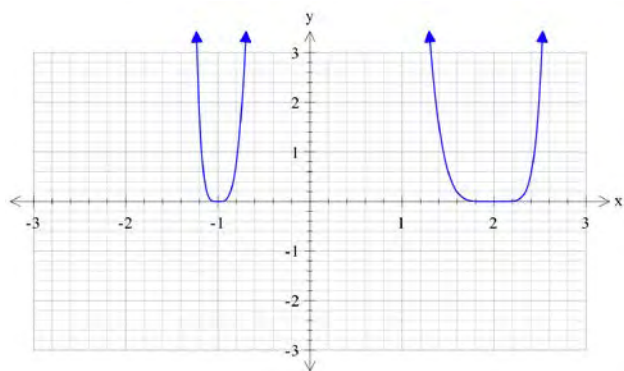
(A)



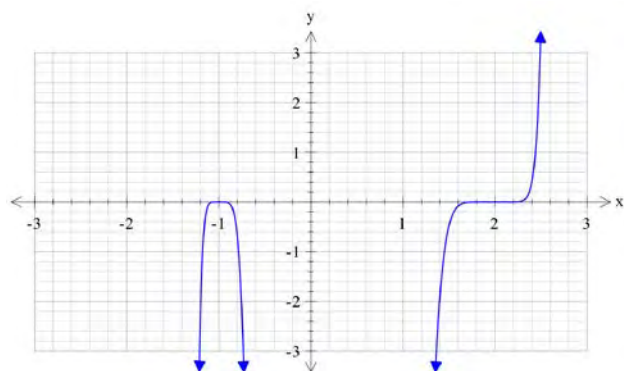
(B)



(C)



(D)



Section II

90 Marks

Attempt Questions 11 - 16.

Allow about 2 hours and 45 minutes for this section.

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available. In Questions 11 - 16, your responses should include relevant mathematics reasoning and/ or calculations.

Question 11 (15 MARKS) Use a SEPARATE writing booklet **Marks**

(a) Let $w = \sqrt{3} + i$ and $z = 3 - \sqrt{3}i$.

(i) Find wz **1**

(ii) Express w in modulus-argument form. **2**

(iii) Write w^4 in simplest Cartesian form. **2**

(b) (i) Find the values of A , B , C and D such that: **2**

$$\frac{5x^3 - 3x^2 + 2x - 1}{x^4 + x^2} = \frac{A}{x} + \frac{B}{x^2} + \frac{Cx + D}{x^2 + 1}$$

(ii) Hence find $\int \frac{5x^3 - 3x^2 + 2x - 1}{x^4 + x^2} dx$. **2**

(c) Find all solutions to the equation $x^4 - 2x^3 + x^2 - 8x - 12 = 0$, given that $x = 2i$ is a root of the equation. **3**

(d) Find $\int \frac{x \, dx}{x^2 - 3x + 4}$ **3**

Question 12 (15 marks) Use a SEPARATE writing booklet	Marks
(a) Using calculus, show that $x \geq \ln(1+x)$ for $x \geq -1$.	3
(b) Consider the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$	
(i) Show that the point $P(4 \cos \theta, 3 \sin \theta)$ lies on the ellipse.	1
(ii) Calculate the eccentricity of the ellipse and hence find the foci and the directrices of the ellipse.	3
(iii) Find the equation of the tangent at $P(4 \cos \theta, 3 \sin \theta)$.	2
(iv) Find the equation of the normal at $P(4 \cos \theta, 3 \sin \theta)$.	2
(v) Show that the tangent at P cuts the positive directrix at $M\left(\frac{16\sqrt{7}}{7}, \frac{21-12\sqrt{7}\cos\theta}{7\sin\theta}\right)$.	2
(vi) Hence show that $\angle PSM = 90^\circ$, if S is the positive focus.	2

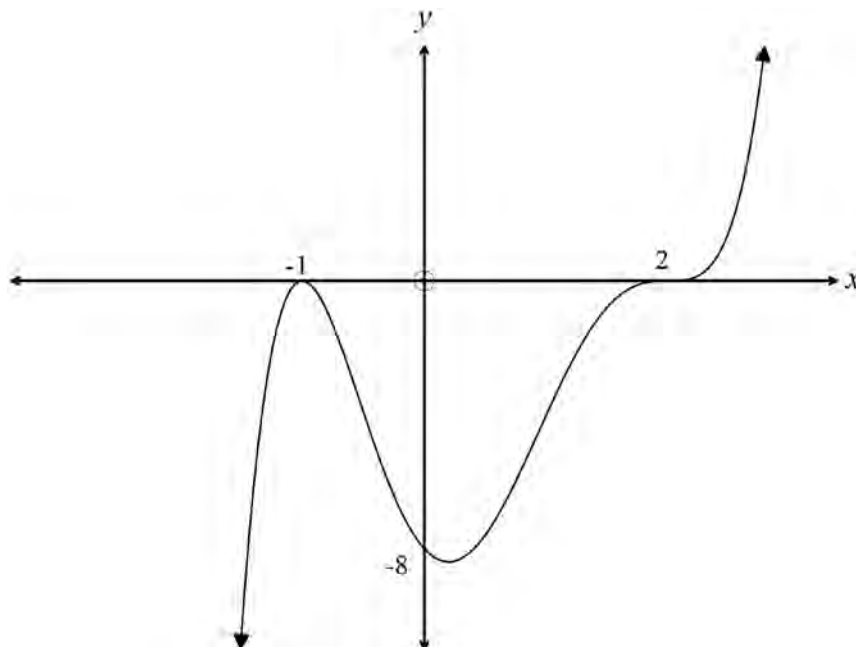
Question 13 (15 marks) Use a SEPARATE writing booklet

Marks

(a) Express $\frac{(1+i)^2}{(1-i\sqrt{3})^2}$ in the form $r \operatorname{cis} \theta$.

4

(b) The graph of $y = f(x)$ is shown below.



Sketch the following curves on separate half page diagrams.

(i) $y = |f(x)|$

1

(ii) $y = \frac{1}{f(x)}$

2

(iii) $y = \frac{d}{dx} [f(x)]$

2

(iv) $y^2 = f(x)$

2

(c) Prove that $\operatorname{cis} (\alpha + \beta) = \operatorname{cis} \alpha \operatorname{cis} \beta$

2

(d) Find the Cartesian equation of the following curve and sketch it on an Argand Diagram.

2

$$|z + 3 + 2i| = |z - 2 + i|$$

Question 14 (15 marks) Use a SEPARATE writing booklet**Marks**

- (a) (i) Let $I_n = \int x(\ln x)^n dx$ for $n = 0, 1, 2, 3, \dots$

3

Show that $I_n = \frac{x^2}{2}(\ln x)^n - \frac{n}{2}I_{n-1}$ for $n \geq 1$

1

- (ii) Hence, or otherwise, find $\int x(\ln x)^2 dx$.

- (b) If $T_1 = 8$, $T_2 = 20$ and $T_n = 4T_{n-1} - 4T_{n-2}$ for $n \geq 3$ prove by mathematical induction that:

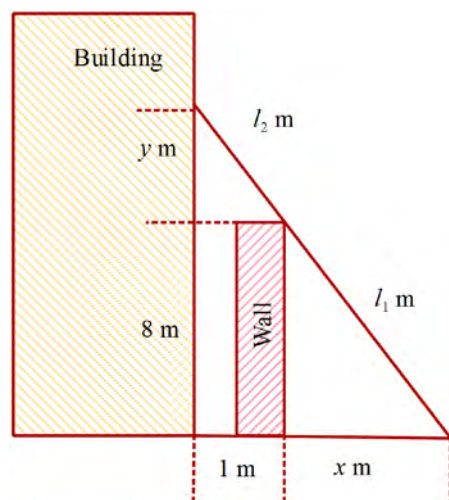
3

$$T_n = (n+3)2^n \text{ for } n \geq 1$$

- (c) Using the method of cylindrical shells, find the volume of the solid of revolution formed when the area bounded by $y = \sin x$, the x -axis, between $x = 0$ and $x = \pi$, is rotated about the y -axis.

3

- (d)



A ladder reaches from the ground, over a wall 8 metres high, to the side of a building 1 metre behind the wall.

- (i) By using similar triangles, show that $y = \frac{8}{x}$ and hence the length l of the ladder, where $l = l_1 + l_2$, is given by:

2

$$l = \sqrt{x^2 + 64} + \sqrt{1 + \frac{64}{x^2}}.$$

- (ii) Hence find the length of the shortest ladder which will satisfy the conditions described above.

3

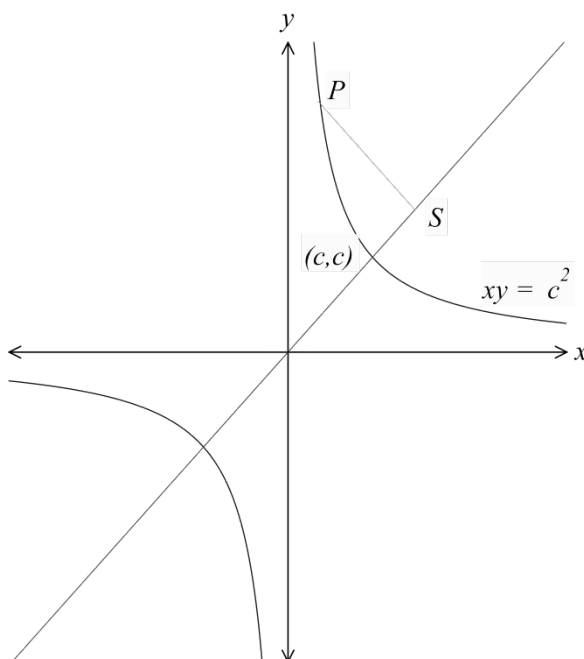
Question 15 (15 marks) Use a SEPARATE writing booklet

Marks

4

(a) Use the substitution $u = e^x$ to find $\int \frac{e^x + e^{2x}}{1 + e^{2x}} dx$.

(b) The point $P\left(cp, \frac{c}{p}\right)$ with $p > 0$ lies on the rectangular hyperbola $xy = c^2$ with focus S . The point T divides the interval PS in the ratio 1:2.



(i) Show that the coordinates of T are:

2

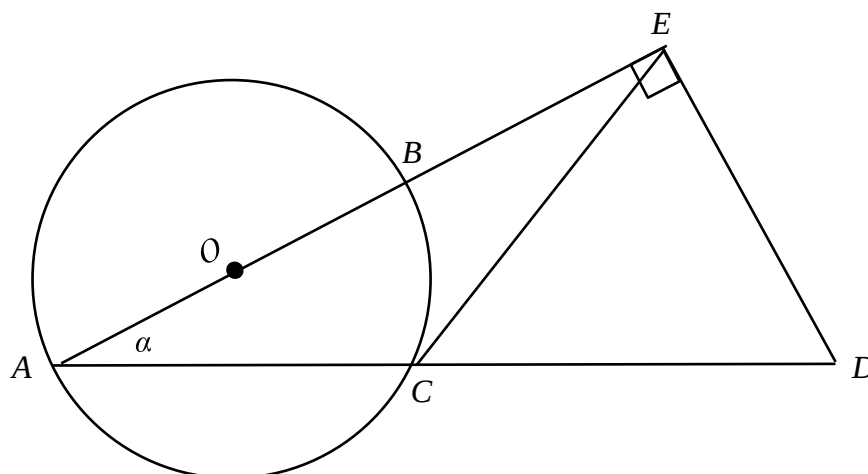
$$T\left(\frac{2cp + c\sqrt{2}}{3}, \frac{\frac{2c}{p} + c\sqrt{2}}{3}\right)$$

(ii) Show that the Cartesian equation of the locus of T can be written as:
 $4c^2 = (3x - c\sqrt{2})(3y - c\sqrt{2})$.

2

HINT: find an expression for $2p$ in terms of x , and $\frac{2}{p}$ in terms of y .

(c)



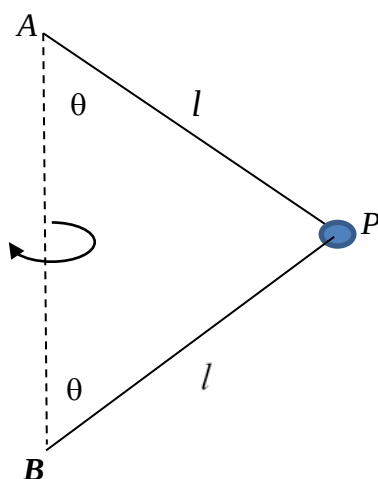
The diameter AB of a circle centre O is produced to E . EC is a tangent touching the circle at C , and the perpendicular to AE at E meets AC produced at D .

3

$$\angle BAC = \alpha$$

Show that $\triangle CDE$ is isosceles.

- (d) A particle P of mass 5 kg is attached by two chains, each of length 3 m , to two fixed points A and B , which lie on a vertical plane. P revolves with constant angular velocity ω about AB . AP makes an angle of θ with the vertical. The tension in AP is T_1 and the tension in BP is T_2 where $T_1 \geq 0$ and $T_2 \geq 0$.



- (i) Resolve the forces on P in the horizontal and vertical directions
- (ii) If the object is rotating in a circle of radius 1.5 m at 12 m/s , find the tension in both parts of the string. (Use $g = 10\text{ m/s}^2$)

2

2

Question 16 (15 marks) Use a SEPARATE writing booklet

Marks

- (a) Find the first derivative of $y = \ln\left(\frac{\sqrt{x^2 + 1}}{\sqrt[3]{x^3 + 1}}\right)$ **2**
- (b) (i) The displacement (from a fixed point) of a body moving in a straight line is given by x , and its velocity is v . **1**
Show that $\frac{d}{dx}\left(\frac{1}{2}v^2\right) = v\frac{dv}{dx}$.
- (ii) A particle of mass one kg is moving in a straight line. It is initially at the origin and is travelling with velocity $\sqrt{3}ms^{-1}$. The particle is moving against a resisting force $v + v^3$, where v is the velocity.
- A Briefly explain why the acceleration of the particle is given by $\frac{dv}{dt} = -(v + v^3)$. **1**
- B Show that the displacement x of the particle from the origin is given by $x = \tan^{-1}\left(\frac{\sqrt{3} - v}{1 + v\sqrt{3}}\right)$. **4**
- C Show that the time t which has elapsed when the particle is travelling with velocity V is given by $t = \frac{1}{2}\log_e\left[\frac{3(1 + V^2)}{4V^2}\right]$ **4**
- D Find V^2 as a function of t . **2**
- E Hence find the limiting position of the particle as $t \rightarrow \infty$. **1**

End of Examination

Mathematics Extension 2

Section I Multiple-Choice Answer Sheet

- | | | | | |
|----|-------------------------|-------------------------|-------------------------|-------------------------|
| 1 | A ☐ | B ☐ | C ☐ | D ☐ |
| 2 | A ☐ | B ☐ | C ☐ | D ☐ |
| 3 | A ☐ | B ☐ | C ☐ | D ☐ |
| 4 | A ☐ | B ☐ | C ☐ | D ☐ |
| 5 | A ☐ | B ☐ | C ☐ | D ☐ |
| 6 | A ☐ | B ☐ | C ☐ | D ☐ |
| 7 | A ☐ | B ☐ | C ☐ | D ☐ |
| 8 | A ☐ | B ☐ | C ☐ | D ☐ |
| 9 | A ☐ | B ☐ | C ☐ | D ☐ |
| 10 | A ☐ | B ☐ | C ☐ | D ☐ |

MULTIPLE CHOICE			
1	$v = \frac{48}{3} = 16m/s$ $v = rw$ $16 = 4w$ $w = \frac{16}{4} = 4 \text{ rads / s}$		B
2	$x^2 - 2x + y^2 = 9$ Differentiate w.r.t x $2x - 2 + 2y \frac{dy}{dx} = 0$ $\frac{dy}{dx} = \frac{1-x}{y}$ At $(0, -3)$, $\frac{dy}{dx} = -\frac{1}{3}$	A	1
3	First person can select 2 out of 6 i.e. $\binom{6}{2}$ ways. Second person can select 2 out of 4 i.e. $\binom{4}{2}$ ways. Final person can chose remaining 2 in 1 way. Number of ways = $\binom{6}{2} \times \binom{4}{2} \times 1 = 15 \times 6 \times 1 = 90$ ways.		C
4	$\int \sin^3 x dx = \int \sin x (\sin^2 x) dx$ $= \int \sin x (1 - \cos^2 x) dx$ $= \int \sin x dx - \int \sin x \cos^2 x dx$ $= -\cos x + \frac{1}{3} \cos^3 x + C = \frac{1}{3} \cos^3 x - \cos x + C$		1 Mark: A
5	Resolving the forces vertically and horizontally at P $T \cos \theta - mg = 0$ $T \sin \theta = mr\omega^2$ Statement (B) is correct.		1 Mark: B
6	$x = \alpha, \beta, \gamma$ so $y = \frac{2}{\alpha}, \frac{2}{\beta}, \frac{2}{\gamma}$. $\therefore y = \frac{2}{x}$ and hence $x = \frac{2}{y}$ $3\left(\frac{2}{y}\right)^3 - 2\left(\frac{2}{y}\right)^2 + \left(\frac{2}{y}\right) - 7 = 0$ $\left(\frac{24}{y^3}\right) - \left(\frac{8}{y^2}\right) + \frac{2}{y} - 7 = 0$ $24 - 8y + 2y^2 - 7y^3 = 0$ Required equation is: $7x^3 - 2x^2 + 8x - 24 = 0$		B

7	$xy = c$ $y + x \frac{dy}{dx} = 0$ $\frac{dy}{dx} = -\frac{y}{x}$ $\frac{dy}{dx} = -\frac{c}{p} \div cp$ $= -\frac{c}{cp^2}$ $= -\frac{1}{p^2}$ $\therefore \text{Gradient of Normal} = p^2$ $y - y_1 = m(x - x_1)$ $y - \frac{c}{p} = p^2(x - cp)$ $py - c = p^3(x - cp)$	D	
8	$\frac{y^2}{225} - \frac{x^2}{64} = 1, a^2 = 64 \text{ and } b^2 = 225$ $a^2 = b^2(e^2 - 1)$ $64 = 225 \times (e^2 - 1)$ $e = \sqrt{\frac{64}{225} + 1} = \sqrt{\frac{289}{225}} = \frac{17}{15}$	1 Mark: C	
9	$z = 2 + 3i$ $i\bar{z} = i(2 + 3i)$ $= i(2 - 3i)$ $= 3 + 2i$	D	
10	Graph (C)	C	

QUESTION 11

a)	$w = \sqrt{3} + i$ and $z = 3 - \sqrt{3}i$. (i) wz $= (\sqrt{3} + i)(3 - \sqrt{3}i)$ $= 3\sqrt{3} - 3i + 3i + \sqrt{3}$ $= 3\sqrt{3} + \sqrt{3}$ $= 4\sqrt{3}$ (ii) $r = \sqrt{(\sqrt{3})^2 + 1^2} = 2$ $\tan \theta = \frac{1}{\sqrt{3}}, \theta = \frac{\pi}{6}$ $\therefore w = 2 \operatorname{cis} \frac{\pi}{6}$ (iii) $w^4 = \left(2 \operatorname{cis} \frac{\pi}{6}\right)^4$ $= 2^4 \operatorname{cis} \frac{4\pi}{6}$ $= 16 \operatorname{cis} \frac{2\pi}{3}$ $= 16 \left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}\right)$ $= 16 \left(-\frac{1}{2} + \frac{\sqrt{3}i}{2}\right)$ $= -8 + 8\sqrt{3}i$	1	Correct Answer
		2	1 for correct r 1 for correct θ
		2	1 – Evaluating Power 1 Answer in Cartesian form
b)	(i) $\frac{5x^3 - 3x^2 + 2x - 1}{x^4 + x^2} = \frac{A}{x} + \frac{B}{x^2} + \frac{Cx + D}{x^2 + 1}$ $5x^3 - 3x^2 + 2x - 1 \equiv Ax(x^2 + 1) + B(x^2 + 1) + (Cx + D)x^2$ $\equiv Ax^3 + Ax + Bx^2 + B + Cx^3 + Dx^2$ $(A + C)x^3 = 5x^3 \quad \therefore A + C = 5$ $(B + D)x^2 = -3x^2 \quad \therefore B + D = -3$ $Ax = 2 \quad \therefore A = 2 \quad \therefore C = 3$ $B = -1 \quad \therefore D = -2$ Hence, $A = 2, B = -1, C = 3, D = -2$. (ii) $\int \frac{5x^3 - 3x^2 + 2x - 1}{x^4 + x^2} dx = \int \left(\frac{2}{x} - \frac{1}{x^2} + \frac{3x - 2}{x^2 + 1}\right) dx$ $= \int \left(\frac{2}{x} - \frac{1}{x^2} + \frac{3x}{x^2 + 1} - \frac{2}{x^2 + 1}\right) dx$ $= 2 \ln x + \frac{1}{x} + \frac{3}{2} \ln(x^2 + 1) - 2 \tan^{-1} x + c$	2	2 – Correct A, B, C and D 1 – 3 correct
		2	1 – Breakup of Integral 1 – Correct Answer
c)	$x^4 - 2x^3 + x^2 - 8x - 12$ Since $x = 2i$ is one root, $(x - 2i)(x + 2i)$ are factors as coeffs are real, so $(x^2 + 4)$ is a factor By division, $x^4 - 2x^3 + x^2 - 8x - 12 = (x^2 + 4)(x^2 - 2x - 3)$ $= (x + 2i)(x - 2i)(x - 3)(x + 1)$ $\therefore$ Solution is $x = \pm 2i, -1$ and 3	3	1 obtaining factor 1 – Correct division 1 all 4 roots

2014 Mathematics Extension 2 HSC Trial Examination Solutions

d)	$\int \frac{x}{x^2 - 3x + 4} dx$ $= \frac{1}{2} \int \frac{2x-3}{x^2-3x+4} + \frac{3}{2} \int \frac{1}{x^2-3x+4}$ $= \frac{1}{2} \ln(x^2 - 3x + 4) + \frac{3}{2} \int \frac{1}{\left(x - \frac{3}{2}\right)^2 + \frac{7}{4}}$ $= \frac{1}{2} \ln(x^2 - 3x + 4) + \frac{\frac{3}{2}}{\frac{\sqrt{7}}{2}} \tan^{-1} \left(\frac{x - \frac{3}{2}}{\frac{\sqrt{7}}{2}} \right) + c$ $= \frac{1}{2} \ln(x^2 - 3x + 4) + \frac{3}{\sqrt{7}} \tan^{-1} \left(\frac{2x - 3}{\sqrt{7}} \right) + c$	1 1 1	Total marks - 3
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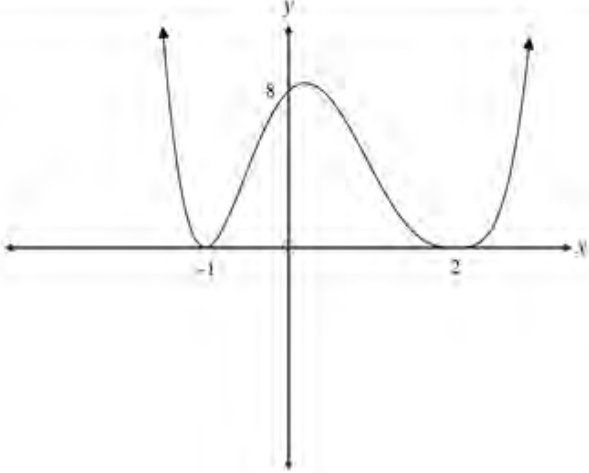
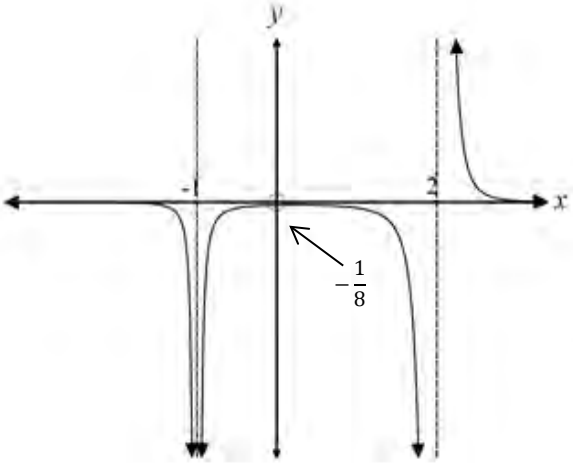
QUESTION 12

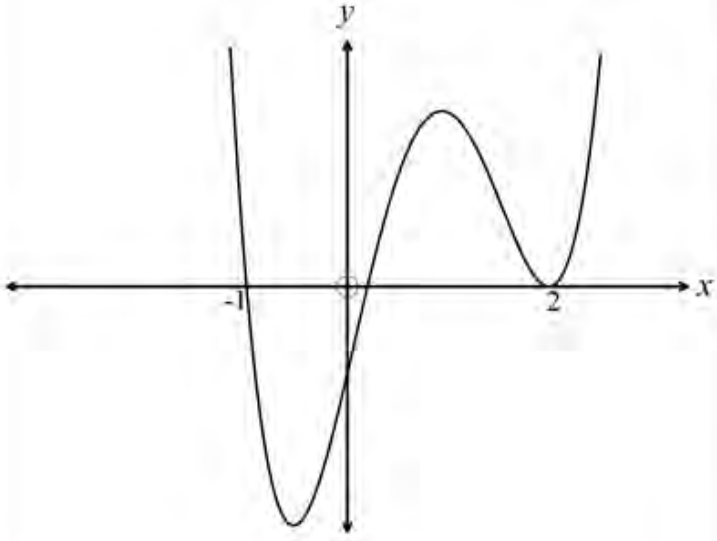
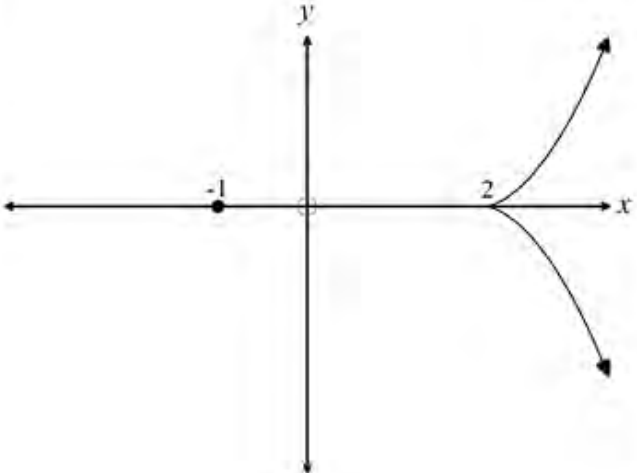
a)	Let $f(x) = x - \ln(1+x)$ and $f'(x) = 1 - \frac{1}{1+x}$ Minimum occurs if $f'(x) = 0$ $1 - \frac{1}{1+x} = 0 \quad \frac{1+x}{1+x} - \frac{1}{1+x} = 0$ $\therefore 1+x-1=0 \text{ or } x=0 \quad (x \neq -1)$ Test $f''(x) = \frac{1}{(1+x)^2}$, $f''(0) = 1 > 0$ Minima Therefore the least value of $f(x)$ is at $x=0$ $f(0) = 0 - \ln(1+0) = 0 \text{ hence } f(x) \geq 0$ $f(x) = x - \ln(1+x) \geq 0$ $\therefore x \geq \ln(1+x)$	3 Marks: Correct answer. 2 Marks: Makes significant progress towards the solution. 2 Marks: 1 Mark: Sets up the function and correctly uses calculus.	
b)	(i) $\frac{x^2}{16} + \frac{y^2}{9} = 1$ $P(4\cos \theta, 3\sin \theta)$ $\frac{(4\cos \theta)^2}{16} + \frac{(3\sin \theta)^2}{9} = 1$ $\frac{16\cos^2 \theta}{16} + \frac{9\sin^2 \theta}{9} = 1$ $\cos^2 \theta + \sin^2 \theta = 1$ $1 = 1$ $\therefore P$ lies on the ellipse. (ii) $b^2 = a^2(1 - e^2)$ $3^2 = 4^2(1 - e^2)$ $\frac{9}{16} = 1 - e^2$ $e^2 = 1 - \frac{9}{16}$ $e^2 = \frac{7}{16}$ $e = \frac{\sqrt{7}}{4}$ Foci $= (\pm ae, 0) = \left(\pm 4 \times \frac{\sqrt{7}}{4}, 0\right) = (\pm\sqrt{7}, 0)$ Directrices : $x = \pm \frac{a}{e}$ $x = \pm 4 \div \frac{\sqrt{7}}{4}$ $x = \pm \frac{16}{\sqrt{7}} = \pm \frac{16\sqrt{7}}{7}$	1 Working 3 1 – eccentricity 1 – foci 1 – <u>directrices</u>	
	(iii) If $\frac{x^2}{16} + \frac{y^2}{9} = 1$ $\frac{2x}{16} + \frac{2y}{9} \frac{dy}{dx} = 0$ $\frac{dx}{dy} = -\frac{16}{2x} \div \frac{2y}{9}$ $\frac{dx}{dy} = -\frac{16}{2x} \times \frac{9}{2y}$ $\frac{dy}{dx} = -\frac{9x}{16y}$ At $P(4\cos \theta, 3\sin \theta)$ $\frac{dy}{dx} = -\frac{36\cos \theta}{48\sin \theta} = -\frac{3\cos \theta}{4\sin \theta}$ $y - y_1 = m(x - x_1)$ $y - 3\sin \theta = -\frac{3\cos \theta}{4\sin \theta}(x - 4\cos \theta)$ $4y\sin \theta - 12\sin^2 \theta = -3x\cos \theta + 12\cos^2 \theta$ $3x\cos \theta + 4y\sin \theta = 12$	2 1 – gradient 1 – Equation	

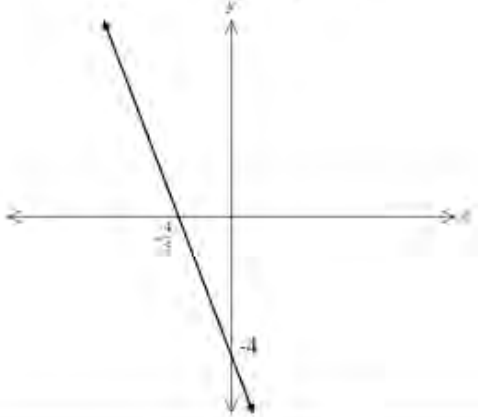
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	(iv) Normal $\frac{dy}{dx} = \frac{4 \sin \theta}{3 \cos \theta}$ $y - y_1 = m(x - x_1)$ $y - 3 \sin \theta = \frac{4 \sin \theta}{3 \cos \theta} (x - 4 \cos \theta)$ $3y \cos \theta - 9 \sin \theta \cos \theta = 4x \sin \theta - 16 \sin \theta \cos \theta$ $4x \sin \theta - 3y \cos \theta - 7 \sin \theta \cos \theta = 0$	2	1 – substitution 1 – answer
	(v) $3x \cos \theta + 4y \sin \theta = 12$ $x = \frac{16\sqrt{7}}{7}$ $\frac{48\sqrt{7}}{7} \cos \theta + 4y \sin \theta = 12$ $4y \sin \theta = 12 - \frac{48\sqrt{7}}{7} \cos \theta$ $y = \frac{12 - \frac{48\sqrt{7}}{7} \cos \theta}{4 \sin \theta}$ $y = \frac{84 - 48\sqrt{7} \cos \theta}{28 \sin \theta} = \frac{21 - 12\sqrt{7} \cos \theta}{7 \sin \theta}$ $\text{Therefore } M = \left(\frac{16\sqrt{7}}{7}, \frac{21 - 12\sqrt{7} \cos \theta}{7 \sin \theta} \right)$	2	1 – substitution 1 – working
	(vi) $\text{Gradient PS} = \frac{3 \sin \theta - 0}{4 \cos \theta - \sqrt{7}} = \frac{3 \sin \theta}{4 \cos \theta - \sqrt{7}}$ $\text{Gradient MS} = \frac{\frac{21 - 12\sqrt{7} \cos \theta}{7 \sin \theta} - 0}{\frac{16\sqrt{7}}{7} - \sqrt{7}} = \frac{\frac{21 - 12\sqrt{7} \cos \theta}{7 \sin \theta}}{\frac{16\sqrt{7} - 7\sqrt{7}}{7}} = \frac{\frac{21 - 12\sqrt{7} \cos \theta}{7 \sin \theta}}{\frac{9\sqrt{7}}{7}} = \frac{21 - 12\sqrt{7} \cos \theta}{9\sqrt{7} \sin \theta}$ $m(\text{PS}) \cdot m(\text{MS}) = \frac{3 \sin \theta}{4 \cos \theta - \sqrt{7}} \cdot \frac{21 - 12\sqrt{7} \cos \theta}{9\sqrt{7} \sin \theta}$ $= \frac{7 - 4\sqrt{7} \cos \theta}{(4 \cos \theta - \sqrt{7})\sqrt{7}}$ $= \frac{7 - 4\sqrt{7} \cos \theta}{4\sqrt{7} \cos \theta - 7}$ $= \frac{7 - 4\sqrt{7} \cos \theta}{-(7 - 4\sqrt{7} \cos \theta)}$ $= -1$ $\therefore MS \perp PS$ and $\angle PSM = 90^\circ$	2	1 – Both Gradients 1 – proving perpendicular

QUESTION 13

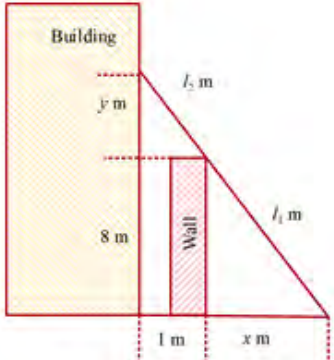
a)	$\frac{(1+i)^2}{(1-i\sqrt{3})^2}$ $(1+i): r = \sqrt{2}, \theta = \frac{\pi}{4} \quad 1+i = \sqrt{2} \operatorname{cis} \frac{\pi}{4}$ $(1-i\sqrt{3}): r = 2, \theta = -\frac{\pi}{3} \text{ so } 1-i\sqrt{3} = 2 \operatorname{cis} \left(-\frac{\pi}{3}\right)$ $\frac{(1+i)^2}{(1-i\sqrt{3})^2} = \frac{\left(\sqrt{2} \operatorname{cis} \frac{\pi}{4}\right)^2}{\left(2 \operatorname{cis} \left(-\frac{\pi}{3}\right)\right)^2}$ $= \frac{2 \operatorname{cis} \frac{\pi}{2}}{4 \operatorname{cis} \left(-\frac{2\pi}{3}\right)}$ $= \frac{1}{2} \operatorname{cis} \left(\frac{7\pi}{6}\right)$ $= \frac{1}{2} \operatorname{cis} \left(\frac{-5\pi}{6}\right)$	1 1 1 1	2 marks for converting to mod-arg form 2 marks for any valid method of simplifying.
b)	(i)  (ii) 	1 2	Correct Graph 1- Shape of Graph 1- Accuracy of critical points

	(iii) 	2	1- Shape of Graph 1 – Accuracy of critical points
	(iv) 	2	1- Shape of Graph 1 – Accuracy of critical points
c)	$\text{cis}(\alpha + \beta) = \text{cis } \alpha \text{ cis } \beta$ $\text{LHS} = \cos(\alpha + \beta) + i \sin(\alpha + \beta)$ $\quad = \cos \alpha \cos \beta - \sin \alpha \sin \beta + i(\sin \alpha \cos \beta + \cos \alpha \sin \beta)$ $\text{RHS} = \text{cis } \alpha \text{ cis } \beta$ $\quad = (\cos \alpha + i \sin \alpha)(\cos \beta + i \sin \beta)$ $\quad = \cos \alpha \cos \beta + i \sin \beta \cos \alpha + i \sin \alpha \cos \beta - \sin \alpha \sin \beta$ $\quad = \cos \alpha \cos \beta - \sin \alpha \sin \beta + i(\sin \alpha \cos \beta + \cos \alpha \sin \beta)$ $\quad = \text{LHS}$	1 1	LHS RHS

d)	$ z + 3 + 2i = z - 2 + i $ $(x + 3)^2 + (y + 2)^2 = (x - 2)^2 + (y + 1)^2$ $x^2 + 6x + 9 + y^2 + 4y + 4 = x^2 - 4x + 4 + y^2 + 2y + 1$ $10x + 2y + 8 = 0$ $5x + y + 4 = 0$ $y = -5x - 4$ 	1	1 - equation
		1	1 - Graph

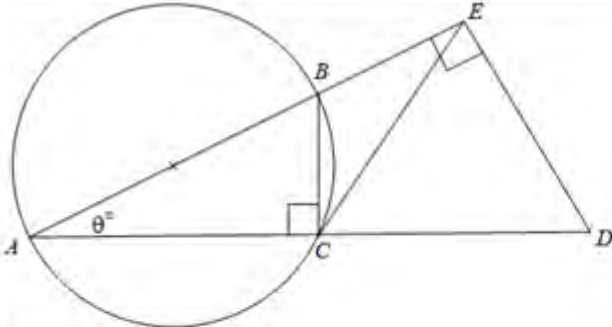
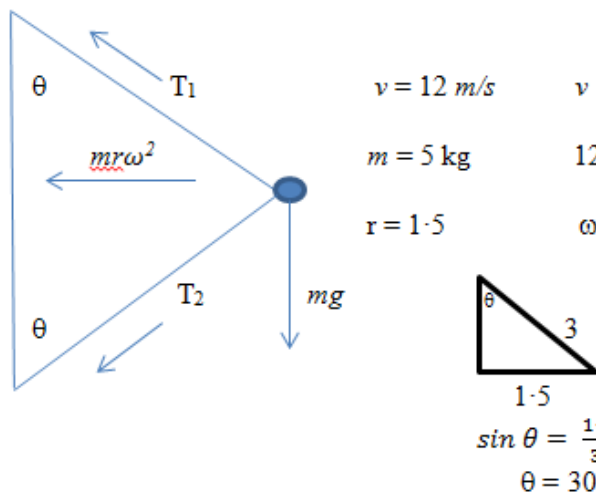
QUESTION 14

a)	$I_n = \int x(\ln x)^n dx$ $= (\ln x)^n \frac{x^2}{2} - \int \frac{x^2}{2} \times \frac{n}{x} (\ln x)^{n-1} dx$ $= (\ln x)^n \frac{x^2}{2} - \frac{n}{2} \int x(\ln x)^{n-1} dx$ $= (\ln x)^n \frac{x^2}{2} - \frac{n}{2} I_{n-1} \text{ for } n \geq 1$	3 Marks: Correct answer. 2 Marks: Makes significant progress towards the solution. 1 Mark: Sets up the integration and shows some understanding. by parts
	$I_2 = \int x(\ln x)^2 dx = \frac{x^2(\ln x)^2}{2} - \frac{x^2 \ln x}{2} - \frac{x^2}{4} + C$	1 mark correct
b)	Step 1: To prove the statement true for $n = 1$ and $n = 2$ $T_1 = (1+3)2^1 = 8 \quad T_2 = (2+3)2^2 = 20$ Result is true for $n = 1$ Result is true for $n = 2$ Step 2: Assume the result true for $n = k$, $n = k-1$ $T_k = (k+3)2^k \quad , \quad T_{k-1} = (k+2)2^{k-1}$ To prove the result is true for, $T_{k+1} = (n+4)2^{k+1}$ $T_{k+1} = 4T_k - 4T_{k-1}$ $= 4(k+3)2^k - 4(k+2)2^{k-1} \quad \text{Using assumption in 2}$ $= 4k2^k + 12 \times 2^k - 4k2^{k-1} - 8 \times 2^{k-1}$ $= 4k2^k + 12 \times 2^k - 2k2^k - 4 \times 2^k$ $= 2^{k+1}(2k+6-k-2)$ $= (k+4)2^{k+1}$ Result is true for $n = k+1$ if true for $n = k$ Step 3: Result true by principle of mathematical induction.	3 Marks: Correct answer. 2 Marks: Proves the result true for $n = 1$ and attempts to use the result of $n = k$ to prove the result for $n = k+1$. 1 Mark: Proves the result true for $n = 1$ and $n = 2$.
c)	By shell Method $\lim_{\delta x \rightarrow 0} 2\pi x \sin x \delta x$ $V = 2\pi \int_0^\pi x \sin x dx$ let $u = x$ $v' = \sin x$ $u' = 1$ $v = -\cos x$ $I = uv - \int vu'$ $= 2\pi [-x \cos x]_0^\pi - \int_0^\pi -\cos x dx$ $= 2\pi [\pi + (\sin x)_0^\pi]$ $= 2\pi^2$	1 Shell Method 1 Integration by parts 1 Answer

d) (i)	 $\frac{y}{8} = \frac{1}{x} \Rightarrow y = \frac{8}{x}$ $l_1^2 = x^2 + 64 \rightarrow l_1 = \sqrt{x^2 + 64}$ $l_2^2 = y^2 + 1 \rightarrow l_2 = \sqrt{y^2 + 1}$ $l = \sqrt{x^2 + 64} + \sqrt{y^2 + 1}$ $l = (x^2 + 64)^{\frac{1}{2}} + \left(\frac{64}{x^2} + 1\right)^{\frac{1}{2}}$ $l = (x^2 + 64)^{\frac{1}{2}} + (1 + 64x^{-2})^{\frac{1}{2}}$	1 Equations for l_1 and l_2 1 Equation for l
(ii)	$l' = \frac{1}{2}(2x)(x^2 + 64)^{-\frac{1}{2}} + \frac{1}{2}(-128x^{-3})(1 + 64x^{-2})^{-\frac{1}{2}}$ $l' = \frac{2x}{2\sqrt{x^2 + 64}} - \frac{128}{2x^3 \sqrt{1 + \frac{64}{x^2}}}$ $l' = \frac{x}{\sqrt{x^2 + 64}} - \frac{64}{x^3 \sqrt{\frac{x^2 + 64}{x^2}}}$ $l' = \frac{x}{\sqrt{x^2 + 64}} - \frac{64}{x^2 \sqrt{x^2 + 64}}$	1 Correct derivative
	Stat pt: $l' = \frac{x}{\sqrt{x^2 + 64}} - \frac{64}{x^2 \sqrt{x^2 + 64}} = 0$ $\frac{x}{\sqrt{x^2 + 64}} = \frac{64}{x^2 \sqrt{x^2 + 64}}$	
	$x^3 = \frac{64\sqrt{x^2 + 64}}{\sqrt{x^2 + 64}}$ $x^3 = 64$ $x = 4$ When $x = 1, l' < 0$ When $x = 5, l' > 0$ $\therefore$ minimum when $x = 4$ hence $y = 2$ $l_1 = \sqrt{80} = 4\sqrt{5}, l_2 = \sqrt{5}$ $\therefore L = 4\sqrt{5} + \sqrt{5} = 5\sqrt{5} \approx 11.2$ metres	1 Value of x and test 1 Minimum length

QUESTION 15

a)	$u = e^x \text{ or } du = e^x dx \text{ or } dx = \frac{1}{u} du$ $\int \frac{e^x + e^{2x}}{1 + e^{2x}} dx = \int \frac{u + u^2}{1 + u^2} \times \frac{1}{u} du$ $= \int \frac{1+u}{1+u^2} du$ $= \int \frac{1}{1+u^2} du + \int \frac{u}{1+u^2} du$ $= \tan^{-1} u + \frac{1}{2} \ln(u^2 + 1) + C$ $= \tan^{-1} e^x + \frac{1}{2} \ln(e^{2x} + 1) + C$	4 Marks: Correct answer. 3 Marks: Separates and integrates one part correctly. 2 Marks: Correctly expresses the integral in terms of u 1 Mark: Correctly finds dx in terms of du
b) (i)	$P\left(cp, \frac{c}{p}\right), S(c\sqrt{2}, c\sqrt{2})$ and $PT:TS = 1:2$ Coordinates of T $x = \frac{mx_2 + nx_1}{m+n} \qquad y = \frac{my_2 + ny_1}{m+n}$ $= \frac{1 \times c\sqrt{2} + 2 \times cp}{1+2} \qquad = \frac{1 \times c\sqrt{2} + 2 \times \frac{c}{p}}{1+2}$ $= \frac{c(\sqrt{2} + 2p)}{3} \qquad = \frac{c\left(\sqrt{2} + \frac{2}{p}\right)}{3}$	2 Marks: Correct answer. 1 Mark: Finds one of the coordinates or makes some progress towards the solution.
(ii)	To find the locus of T eliminate p from the above equations. $x = \frac{c(\sqrt{2} + 2p)}{3} \qquad y = \frac{c\left(\sqrt{2} + \frac{2}{p}\right)}{3}$ $3x = c(\sqrt{2} + 2p) \qquad \frac{3y}{c} - \sqrt{2} = \frac{2}{p}$ $\frac{3x}{c} - \sqrt{2} = 2p \qquad \frac{3y}{c} - \sqrt{2} = \frac{2}{p}$ $\left(\frac{3x}{c} - \sqrt{2}\right) \times \left(\frac{3y}{c} - \sqrt{2}\right) = 2p \times \frac{2}{p} = 4$ $(3x - \sqrt{2}c)(3y - \sqrt{2}c) = 4c^2$	2 Marks: Correct answer. 1 Mark: Uses the coordinates of T and attempts to eliminate p.

c)	 Let $\angle BAC = \theta$ $\angle ACB = 90^\circ$ (angle in a semi-circle) $\angle BCD = 90^\circ$ (adjacent angles on a straight line) $\angle BCE = \angle BAC$ (angle between a tangent and a chord equals the angle in the alternate segment) $\therefore \angle BCE = \theta$ $\therefore \angle ECD = 90 - \theta$ (angle sum of $\angle BCD$) $\therefore \angle EDC = 90 - \theta$ ((angle sum of $\triangle AED$) $\therefore \angle ECD = \angle EDC = 90 - \theta$ $\triangle ECD$ is isosceles (base angles of the triangle are equal)	3 Marks: Correct answer. 2 Marks: Makes significant progress towards the solution. 1 Mark: Draws the diagram and applies a relevant circle theorem.
d)	 $v = 12 \text{ m/s}$ $v = r\omega$ $m = 5 \text{ kg}$ $12 = 1.5\omega$ $r = 1.5$ $\omega = 8$ $\sin \theta = \frac{1.5}{3}$ $\theta = 30^\circ$	
	$T_1 \cos \theta = mg + T_2 \cos \theta$ $T_1 \cos 30 = (5)(10) + T_2 \cos 30$ $T_1 \cdot \frac{\sqrt{3}}{2} = 50 + T_2 \cdot \frac{\sqrt{3}}{2}$ $T_1 \sqrt{3} = 100 + T_2 \sqrt{3} \dots \dots (1)$	(can be without numerical sub) 1 Vertical Equation
	$mr\omega^2 = T_1 \sin \theta + T_2 \sin \theta$ $5 \times 1.5 \times 8^2 = T_1 \sin 30 + T_2 \sin 30$ $480 = \frac{T_1}{2} + \frac{T_2}{2}$ $960 = T_1 + T_2 \dots \dots (2)$	(can be without numerical sub) 1 Horizontal Equation

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	From (2), $T_1 = 960 - T_2 \dots \dots (3)$ (3) in (1) $(960 - T_2)\sqrt{3} = 100 + T_2\sqrt{3}$ $960\sqrt{3} - T_2\sqrt{3} = 100 + T_2\sqrt{3}$ $T_2 2\sqrt{3} = 960\sqrt{3} - 100$ $T_2 = \frac{960\sqrt{3} - 100}{2\sqrt{3}} = 451 \text{ N}$ $T_1 = 960 - T_2$ $T_1 = 960 - 451$ $T_1 = 509$	1	Value for T_2
		1	Value for T_1

QUESTION 16			
a)	$y = \ln \left(\frac{\sqrt{x^2 + 1}}{\sqrt[3]{x^3 + 1}} \right)$ $y = \ln \sqrt{x^2 + 1} - \ln \sqrt[3]{x^3 + 1}$ $y = \ln(x^2 + 1)^{\frac{1}{2}} - \ln(x^3 + 1)^{\frac{1}{3}}$ $y = \frac{1}{2} \ln(x^2 + 1) - \frac{1}{3} \ln(x^3 + 1)$ $\frac{dy}{dx} = \frac{1}{2} \frac{2x}{x^2 + 1} - \frac{1}{3} \frac{3x^2}{x^3 + 1}$ $= \frac{x}{x^2 + 1} - \frac{x^2}{x^3 + 1}$	1	Use of logarithmic rules
		1	Answer
b)		1 mark correctly establishes result	
(i)	$\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = \frac{dv}{dx} \times \frac{d}{dv} \left(\frac{1}{2} v^2 \right)$ $= v \frac{dv}{dx}$		
(ii)	Resistance force acts against the direction of motion, $\therefore F = m \times a = 1 \times a = -(v + v^3)$ $a = -(v + v^3)$ $\frac{dv}{dt} = -(v + v^3)$	1 mark correct explanation	
A			
B	$v \frac{dv}{dx} = -(v + v^3)$ $\frac{dv}{dx} = -(1 + v^2)$ $\frac{dx}{dv} = \frac{-1}{1 + v^2}$ $x = -\tan^{-1} v + c$ $\text{when } x = 0, v = \sqrt{3} \Rightarrow c = \tan^{-1} \sqrt{3}$ $\therefore x = \tan^{-1} \sqrt{3} - \tan^{-1} v$ $\tan x = \tan(\tan^{-1} \sqrt{3} - \tan^{-1} v)$ $= \frac{\tan(\tan^{-1} \sqrt{3}) - \tan(\tan^{-1} v)}{1 + \tan(\tan^{-1} \sqrt{3}) \times \tan(\tan^{-1} v)}$ $= \frac{\sqrt{3} - v}{1 + v\sqrt{3}}$ $\therefore x = \tan^{-1} \left(\frac{\sqrt{3} - v}{1 + v\sqrt{3}} \right)$	1 mark some correct progress towards result	
		1 mark	$\frac{dx}{dv} = -\frac{1}{1 + v^2}$
		1 mark	$x = \tan^{-1} \sqrt{3} - \tan^{-1} v$
		1 mark	correctly establishes result

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C	$\frac{dv}{dt} = -(v + v^3)$ $\frac{dt}{dv} = -\frac{1}{v + v^3}$ $= \frac{-1}{v(1 + v^2)}$ $= \frac{v}{1 + v^2} - \frac{1}{v}$ $t = \int_{\sqrt{3}}^v \frac{v}{1 + v^2} - \frac{1}{v} dv$ $= \left[\frac{1}{2} \log_e(1 + v^2) - \log_e v \right]_{\sqrt{3}}^v$ $= \frac{1}{2} \left[\log_e(1 + v^2) - 2 \log_e v \right]_{\sqrt{3}}^v$ $= \frac{1}{2} \left[\log_e \frac{1 + v^2}{v^2} \right]_{\sqrt{3}}^v$ $= \frac{1}{2} \left[\log_e \left(\frac{1 + V^2}{V^2} \right) - \log_e \left(\frac{4}{3} \right) \right]$ $= \frac{1}{2} \log_e \left[\frac{3(1 + V^2)}{4V^2} \right]$	1 mark $\frac{dt}{dv} = -1 \frac{1}{v + v^3}$ 1 mark correct partial fractions 1 mark correct integration 1 mark correct result
D	$t = \frac{1}{2} \log_e \left[\frac{3(1 + V^2)}{4V^2} \right]$ $e^{2t} = \frac{3(1 + V^2)}{4V^2}$ $4V^2 e^{2t} = 3 + 3V^2$ $V^2(4e^{2t} - 3) = 3$ $V^2 = \frac{3}{4e^{2t} - 3}$	1 mark $e^{2t} = \frac{3(1 + V^2)}{4V^2}$ 1 mark correct result
E	As $t \rightarrow \infty, v \rightarrow 0$ hence $x \rightarrow \tan^{-1} \sqrt{3}$	1 mark correct both parts